Big Data Analytics Week 2 Lab: Introduction to R

Dom Richards

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This lab session will cover the basics of R. There are no questions - just examples.

Basic Commands

Here are some basic R commands:

```
# Basic arithmetic
2 + 3
```

```
## [1] 5
```

```
5 * 7
```

```
## [1] 35
```

```
10 / 3
```

```
## [1] 3.333333
```

```
2^4
```

```
## [1] 16
```

```
# Assignment operator
x <- 5
y <- 10
x + y
```

```
## [1] 15
```

```
# Check what's in your environment
ls()
```

```
## [1] "x" "y"
```

Assignment Operators: <- vs =

R has two main assignment operators:

```
# Left assignment with <-
a <- 10
b <- 20

# Assignment with =
c <- 30
d <- 40
```

```r
# Both work for basic assignment
print(a)
```

```
## [1] 10
```

```r
print(c)
```

```
## [1] 30
```

What they do

`<-` assignment operator (puts a value into a name).

`=` either assignment or used to match function arguments by name.

Best practice (what most style guides recommend)

- Use `<-` for assignment.
- Use `=` only for function arguments.

Why this convention?

- **Clearer code**: `=` in calls, `<-` elsewhere.
- **Matches common R community style** (Tidyverse, Google).

Working with Vectors

Creating Vectors

```r
# Numeric vectors
numbers <- c(1, 2, 3, 4, 5)
scores <- c(85, 92, 78, 96, 88)

# Character vectors
names <- c("Alice", "Bob", "Charlie", "Diana")
colours <- c("red", "green", "blue")

# Logical vectors
passed <- c(TRUE, TRUE, FALSE, TRUE, TRUE)

# Sequences
seq1 <- 1:10
seq2 <- seq(from = 0, to = 100, by = 10)
rep1 <- rep(c("A", "B"), times = 5)

# Display some examples
print(numbers)
```

```
## [1] 1 2 3 4 5
```

```r
print(seq1)
```

```
##  [1]  1  2  3  4  5  6  7  8  9 10
```

```r
print(rep1)
```

```
##  [1] "A" "B" "A" "B" "A" "B" "A" "B" "A" "B"
```

Vector Operations

```
# Accessing elements
numbers[1]
```

```
## [1] 1
```

```
numbers[c(1, 3, 5)]
```

```
## [1] 1 3 5
```

```
numbers[-1]
```

```
## [1] 2 3 4 5
```

```
# Vector arithmetic
a <- c(1, 2, 3)
b <- c(4, 5, 6)
a + b
```

```
## [1] 5 7 9
```

```
a * b
```

```
## [1]  4 10 18
```

```
a^2
```

```
## [1] 1 4 9
```

```
# Summary statistics
mean(scores)
```

```
## [1] 87.8
```

```
median(scores)
```

```
## [1] 88
```

```
sd(scores)
```

```
## [1] 6.870226
```

```
min(scores)
```

```
## [1] 78
```

```
max(scores)
```

```
## [1] 96
```

```
summary(scores)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    78.0    85.0    88.0    87.8    92.0    96.0
```

Matrices and Arrays

Creating Matrices

```
# Create a matrix
mat1 <- matrix(1:12, nrow = 3, ncol = 4)
mat1
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    4    7   10
## [2,]    2    5    8   11
## [3,]    3    6    9   12
```

```
# Create matrix by binding vectors
vec1 <- c(1, 2, 3)
vec2 <- c(4, 5, 6)
mat2 <- rbind(vec1, vec2)
mat3 <- cbind(vec1, vec2)

print("Row bind result:")
```

```
## [1] "Row bind result:"
```

```
mat2
```

```
##      [,1] [,2] [,3]
## vec1    1    2    3
## vec2    4    5    6
```

```
print("Column bind result:")
```

```
## [1] "Column bind result:"
```

```
mat3
```

```
##      vec1 vec2
## [1,]    1    4
## [2,]    2    5
## [3,]    3    6
```

```
# Matrix with row and column names
sales <- matrix(c(100, 120, 90, 110, 95, 130),
    nrow = 2, ncol = 3,
    dimnames = list(
        c("Q1", "Q2"),
        c("Product A", "Product B", "Product C")
    )
)
sales
```

```
##    Product A Product B Product C
## Q1       100        90        95
## Q2       120       110       130
```

Matrix Operations

```
# Accessing matrix elements
sales[1, 2] # Row 1, Column 2
```

```
## [1] 90
```

```
sales[1, ] # Entire first row
```

```
## Product A Product B Product C
##       100        90        95
```

```
sales[, "Product A"] # Entire first column
```

```
##  Q1  Q2
## 100 120
```

```
# Matrix arithmetic
sales * 2 # Scalar multiplication
```

```
##    Product A Product B Product C
## Q1       200       180       190
## Q2       240       220       260
```

```
rowSums(sales) # Sum each row
```

```
##  Q1  Q2
## 285 360
```

```
colSums(sales) # Sum each column
```

```
## Product A Product B Product C
##       220       200       225
```

```
apply(sales, 1, mean) # Row means
```

```
##  Q1  Q2
##  95 120
```

```
apply(sales, 2, mean) # Column means
```

```
## Product A Product B Product C
##     110.0     100.0     112.5
```

Data Frames

Creating Data Frames

```
# Create a data frame
students <- data.frame(
    name = c("Alice", "Bob", "Charlie", "Diana", "Eve"),
    age = c(20, 22, 21, 23, 20),
    grade = c("A", "B+", "A-", "A", "B"),
    passed = c(TRUE, TRUE, TRUE, TRUE, TRUE),
    stringsAsFactors = FALSE
)

# View the data frame
students
```

```
##      name age grade passed
## 1   Alice  20     A   TRUE
## 2     Bob  22    B+   TRUE
## 3 Charlie  21    A-   TRUE
## 4   Diana  23     A   TRUE
## 5     Eve  20     B   TRUE
```

```
str(students) # Structure
```

```
## 'data.frame':    5 obs. of  4 variables:
##  $ name  : chr  "Alice" "Bob" "Charlie" "Diana" ...
##  $ age   : num  20 22 21 23 20
```

```
##  $ grade : chr  "A" "B+" "A-" "A" ...
##  $ passed: logi  TRUE TRUE TRUE TRUE TRUE
```

```
summary(students) # Summary
```

```
##      name                age          grade              passed
##  Length:5           Min.   :20.0   Length:5           Mode:logical
##  Class :character   1st Qu.:20.0   Class :character   TRUE:5
##  Mode  :character   Median :21.0   Mode  :character
##                     Mean   :21.2
##                     3rd Qu.:22.0
##                     Max.   :23.0
```

```
head(students) # First few rows
```

```
##      name age grade passed
## 1   Alice  20     A   TRUE
## 2     Bob  22    B+   TRUE
## 3 Charlie  21    A-   TRUE
## 4   Diana  23     A   TRUE
## 5     Eve  20     B   TRUE
```

```
tail(students) # Last few rows
```

```
##      name age grade passed
## 1   Alice  20     A   TRUE
## 2     Bob  22    B+   TRUE
## 3 Charlie  21    A-   TRUE
## 4   Diana  23     A   TRUE
## 5     Eve  20     B   TRUE
```